

Harsha

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192412

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Experiment7

Obj 1clc;

clear;

closeall;

antennas=[2481632];%NumberofAntennaElements gain =

10*log10(antennas); % Beamforming Gain (dB) figure

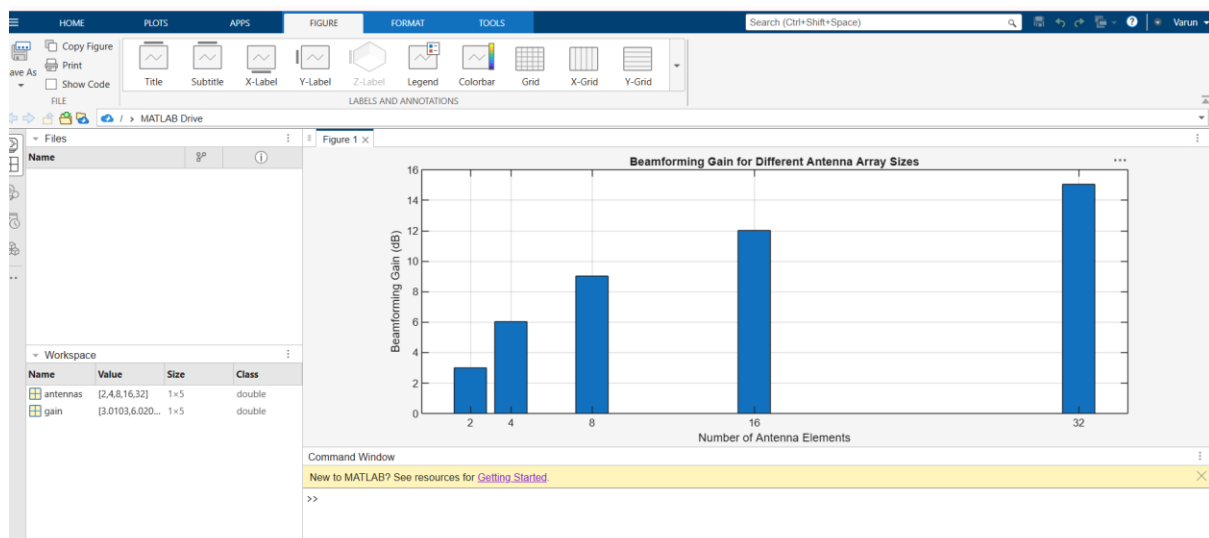
bar(antennas,gain)

grid on

xlabel('NumberofAntennaElements')

ylabel('Beamforming Gain (dB)')

title('BeamformingGainforDifferentAntennaArraySizes')



```

Obj 2clc;

clear;

closeall;

antennas=[2481632];%NumberofAntennaElements

beamwidth=[904522.511.255.625];% BeamWidth(degrees) figure

plot(antennas,beamwidth,'o-','LineWidth',2)

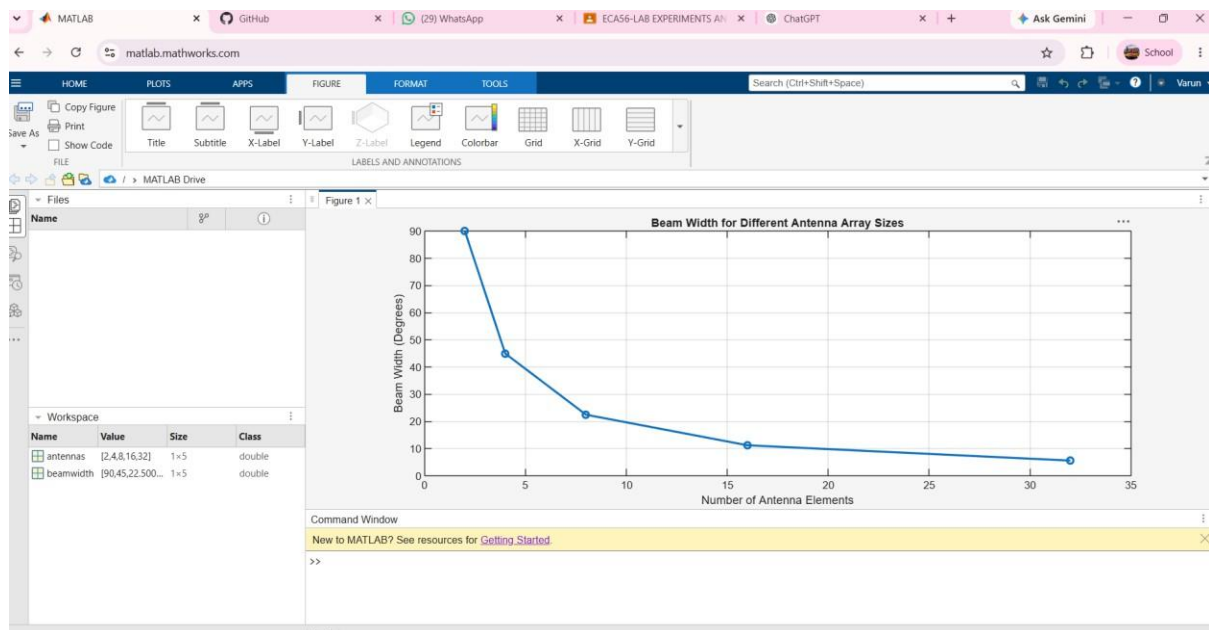
grid on

xlabel('NumberofAntennaElements')

ylabel('Beam Width (Degrees)')

title('BeamWidthforDifferentAntennaArraySizes')

```



```

Obj 3clc;

clear;

closeall;

```

```
theta=[-60-3003060];%MainLobeDirection(degrees)
```

```
gain=[11111];%Constantnormalizedgain figure
```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
```

```
title('MainLobeDirectioninAnalogBeamforming')
```

